Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Console

Selected Domain

Domain 1 - AIVERSECITY

Selected SDG Track (1–17)

4 - Quality Education

Project / Solution Name

ShiftBridge

Name (Leader)

Tan King Yang

Email & Phone (Leader)

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University Affiliation(s)

UTAR

Number of Team Members

1 (solo participant)

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

An experienced manufacturing quality inspector may hear that camera-based inspection is being introduced without knowing which existing skills remain useful or what to learn next. Limited study time makes an unfocused course list a poor answer. Our initial audience is Malaysian electrical and electronics quality inspectors, with training advisers as potential facilitators.

There is evidence for the underlying change. TalentCorp's electrical and electronics workforce study describes evolving QA and process roles and includes automated monitoring among QA skills [S1]. Intel documents computer vision and AI used for inline inspection [S2]. These sources support investigating changed tasks; they do not establish Malaysian job losses or predict one worker's future.

ShiftBridge addresses a narrower decision: given a worker's stated skills, time and budget, which learning action is justified now, and what should change when credible evidence changes? It preserves a route for improving within QA alongside exploratory adjacent roles. This supports access to relevant technical learning under SDG 4.4 and affordable learning under SDG 4.3 [S8].

The specific planning difficulty remains a hypothesis awaiting independent worker or adviser validation. Our prototype uses synthetic profiles and public references. It does not collect real CVs, automate hiring decisions or claim employment, wage or environmental outcomes.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

ShiftBridge connects skills, evidence, role comparison and an affordable learning sequence in one inspectable workflow. A synthetic worker records six skill levels, qualification information, weekly time and budget. Three routes are ranked using deterministic weighted coverage; unknown skills produce a range, and qualification checks remain separate from alignment.
The evidence layer can refresh arXiv research metadata and retains source links, dates and content hashes. A reviewer must identify a supporting passage, explain its task implication, select affected roles and propose bounded scenario adjustments. Approval and publication are separate. Unsupported or irrelevant items leave the accepted plan unchanged.
The runtime AI component is a pinned embedding model for relevance triage on bounded public passages. The hosted app accepted a completed TalentCorp request from the patched RunPod worker; six synthetic smoke cases also completed. Human evidence review remains separate from deterministic planning. Similarity cannot establish deployment, workforce implications or a valid skill rule. A negated-vocabulary case illustrates that limit.
The planner orders learning prerequisites, accounts for known access prices and weekly time, and shows practical outputs. Every role and skill has an SDG explanation. Moderate and faster adoption scenarios expose assumptions rather than promise forecasts. A published change can visibly reorder learning; the worker chooses whether to adopt it. Saved decisions retain the original profile, assessment and evidence version and can be downloaded.
Workshop themes reflected in the architecture include API integration, PostgreSQL persistence and deployment configuration. The participant confirmed completing the GenAI, Database and Cloud Deployment workshop videos. AI-assisted development is disclosed separately from runtime AI. Public hosting and patched GPU execution are verified; representative model evaluation remains outstanding.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Interface: React / TypeScript / Vite; eight responsive pages and a guided rehearsal.

Backend and planning: Node.js / Fastify / Zod; deterministic scoring, qualification checks, prerequisites and explicit worker adoption.

Data: PostgreSQL JSONB on Replit; PGlite locally. The arXiv Atom/RSS connector ingests dated review candidates.

Runtime AI: pinned BAAI/bge-large-en-v1.5 embeddings on RunPod for bounded passage-relevance triage. Human review and separate publication control evidence changes. A hosted v3 request and six synthetic cases completed; accuracy is not established.

Sponsor use - Yes: hackathon-supplied US$20 Replit credits supported hosting/database; US$25 RunPod credits supported GPU inference. No other sponsor or mentor contribution is recorded.

Appendix P contains the system block diagram, full stack details and source citations.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We narrowed the design to one decision: a quality inspector choosing a useful next learning action under time and budget constraints. Public occupational and sector evidence informed the initial taxonomy, while synthetic personas made constraints and unknowns visible. These are design hypotheses, not completed worker interviews.

The prototype allows profile edits, scenario comparison, evidence inspection and an explicit decision to adopt a revision. A constructed rehearsal shows both a relevant update and restraint when evidence is weak. Practice outputs and preserved saved decisions make the result usable beyond a score.

Implementation verification covers scoring, unknowns, prerequisites, review/publication separation, historical preservation and database persistence. Browser checks exercise the real API at desktop and mobile widths. A small independent usability study is planned to compare selecting a justified next step with a static catalogue versus ShiftBridge; no measured improvement is claimed.

The public app passed 24 HTTPS checks; patched v3 hosted inference acceptance and a complete browser demonstration are recorded. A narrated 6:41 walkthrough is rendered; public YouTube publication remains pending. The published sector baseline provides context; independent worker validation and planning-time measurement are future evaluation work, not completed results.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary: SDG 4.4** - relevant technical and vocational skills. **Secondary: SDG 4.3** - access to affordable technical and vocational learning. Route and skill explanations show the intended contribution. These are project-level intentions, not measured UN population indicators. [S8; Appendix P]
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**Published context:** 44 of 50 assessed existing E&E roles (88%) are classified as evolving roles requiring upskilling; Quality Assurance Technician is included [S1, pp. 50-51; Appendix P]. This counts occupation types, not workers or job losses. Selected during proposal completion; no pre-build participant study is claimed. Planning-time baseline: not yet measured (0 participants). The pilot will measure both catalogue and app conditions.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Median planning-task time among successful tasks, together with task success rate, citation correctness, prerequisite errors and help received. Count failures separately. Track automated source-reference and constraint checks as engineering measures. Appendix P gives the pilot method.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**Prospective target: 30% lower median planning time** than the measured static-catalogue condition, without worse task success or source/prerequisite correctness. 100 x (baseline median - app median) / baseline median, when the baseline is positive. Target only; not an achieved result.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

Workshop completion was confirmed by the sole participant via recorded videos. Published QA/upskilling evidence supports the problem context and SDG mapping; independent worker validation remains future work.

The application is deployed and document responses meet their word limits. Public YouTube publication and portal submission remain pending.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

No pre-existing participant-owned project code or design was incorporated in the recorded build. ShiftBridge's project-specific implementation and materials were developed during this hackathon task, beginning 20 October 2026. Public references, pretrained models and third-party packages are declared separately below; they are not claimed as newly created team work.

Codex assistance was substantial in concept development, strategy, design, implementation, verification and documentation. Gemini supplied ideas and critique. The participant worked within the assigned Domain 1, directed scope, revisions and platform use, and authorized the submission preparation. The participant remains responsible for the entry and its claims; the assistants are not the competition organizers or entrants.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React / Fastify / TypeScript	Open-source code; see declaration section 2.2 and Appendix T for all versions.	MIT / Apache-2.0; other package terms
BAAI/bge-large-en-v1.5	Pretrained embedding model. Exact pinned revision and notice in declaration 4.1 / Appendix T.	MIT model notice; no training
TalentCorp / O*NET / Intel	Public occupational and deployment references; attributed excerpts/links.	Publisher-specific rights
arXiv / Microsoft Learn / ASQ / NIST	Public research and learning references; no complete reports or courses redistributed.	Source-specific terms; no blanket relicensing
Synthetic profiles / practice / tests	Original AI-assisted project data; no real worker records.	Apache-2.0 project material

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

Not applicable. The sole participant confirmed that no employer, internship, consulting client or other party has an ownership claim over this project (23 October 2026).

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



Tan King Yang | Team Console | Sole participant / leader

[Date]

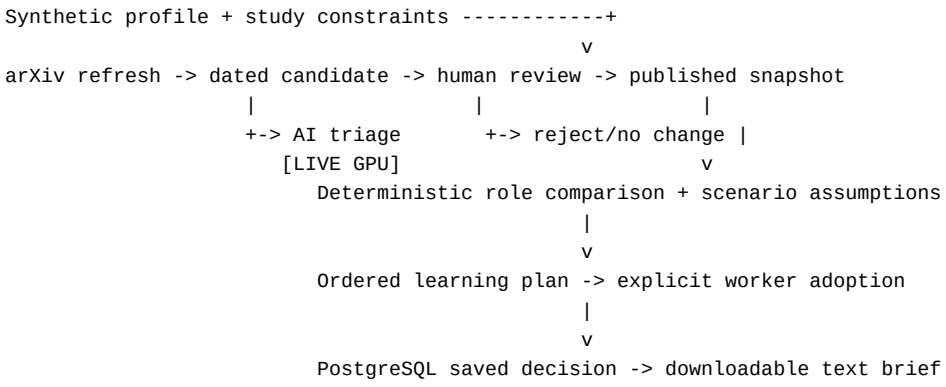
23 October 2026

Official template completed 24 October 2026. Participant-supplied signature and its original date retained with authorization. Codex assisted preparation; the participant is the entrant and signatory.

Appendix P - Architecture, evaluation and references

3.1 Detailed implementation and system block diagram

Layer	Actual implementation and intended use
Interface	React, TypeScript, Vite and Lucide icons; eight responsive application pages plus an isolated guided rehearsal.
API	Node.js and Fastify; Zod validation; separate profile, evaluation, adoption, source-refresh, review and publication operations.
Planning	Shared TypeScript engine 1.1.0; normalized selected-skill coverage, role-scoped evidence, qualification checks, prerequisite-aware starter planning.
Storage	PostgreSQL JSONB workspace records with transactional changes; PGlite for local persistence, pg for managed PostgreSQL. This build does not use a multi-table ORM architecture.
Updatable source	arXiv Atom API with category RSS fallback. Manual refresh is real; a fetched paper is a candidate, not approved labour evidence.
AI integration, initial execution verified	RunPod endpoint and executable worker for BAAI/bge-large-en-v1.5 relevance triage. Model revision is pinned in the contract; the hosted app accepted a v3 TalentCorp result and six declared v3 synthetic cases completed. Wider relevance evaluation remains outstanding.
Hosting, verified public prototype	https://shift-bridge-kingyang3096.replit.app. Published 45bfed5 (r4) passed 24 HTTPS checks; earlier Replit f056cef passed tests/build/smoke.
Verification	Vitest domain/API/connector/client-helper tests, TypeScript checking, production build and recorded browser checks.



Official sponsor API or cloud use: Yes. Replit hosts the public application and managed PostgreSQL. RunPod executed bounded public-passage triage under the current v3 contract. The participant confirmed that the US$20 Replit and US$25 RunPod credits were supplied by the hackathon. No additional sponsor or mentor contribution is recorded for this build. Credits and platform use are disclosed separately from model quality or impact.

4. Evaluation method

use at least five independent participants if feasible; counterbalance two equivalent synthetic cases and condition order. Use the same success rubric: identify a next step, point to its supporting source, respect prerequisites and constraints, and explain one limitation. Record actual timings and failures in pilot_results.csv. Report individual observations and sample size. Small convenience-sample usability results cannot establish employment retention, wage gains or waste reduction. The initial synthetic rehearsal is a software demonstration, not this study.

Sources cited in the official responses

- **[S1]** TalentCorp, Impact Study of Artificial Intelligence, Digital, and Green Economy on the Malaysian Workforce, Volume 2: Electrical and Electronics, copyright 2024, printed pp. 50-51, 67 and 70. Official report. Context, not a particular worker's forecast.

- **[S2]** Intel, Computer Vision and AI for Inline Inspection, December 2023 according to its linked white paper. Deployment account. Date precision is month; it is not a Malaysian employment study.

- **[S3]** O*NET, 51-9061.00, updated 2026. US task reference, not Malaysian labour-market statistics.

- **[S4]** O*NET, 17-3027.00 and 17-3028.00, updated 2026. Exploratory adjacent-role task references, not universal entry rules.

- **[S5]** Microsoft Learn: Power BI, machine-learning concepts and computer-vision concepts. Provider durations are not established; certificates are not included.

- **[S6]** ASQ: Check Sheet and Root Cause Analysis; NIST: Calibration. Public references, not certified four-hour courses.

- **[S7]** arXiv API and cs.CV RSS; PatchCore research example, first submitted 2021-06-15, revised 2022-05-05. Research alone does not prove adoption or altered job duties.

- **[S8]** United Nations, Goal 4 and its targets. The app's planning measures are project proxies, not official SDG indicator measurements.

Catalogue source review: 21 October 2026. Published role-count baseline checked on 23 October 2026. The baseline describes assessed role types, not a measured ShiftBridge outcome.

Final submission links

Console | ShiftBridge | Domain 1 - AIVERSECITY

Publication update: 24 October 2026

This dated administrative update supersedes earlier statements that the YouTube upload is pending. The signed declaration and its original date are preserved without alteration.

Pitch/demo: Console: ShiftBridge - 6 minutes 41 seconds

https://youtu.be/zyS30NtQKsg

Public source and final submission index:

https://github.com/Epsilon3096/SDC_Hackathon

Live prototype:

https://shift-bridge-kingyang3096.replit.app

The participant supplied the uploaded video URL. Browser playback and the 6:41 duration were checked on 24 October 2026. YouTube labels it Unlisted; viewers can access it using the direct link.

The repository README links the slide deck and separate Replit and RunPod proof-of-use PDFs. This sheet records publication readiness; it does not itself certify portal receipt or judging acceptance.